Reflect with us - Week 7

This assignment will not be graded and is only for practice.
1 point
Let L_1 L1 and L_2 L2 be affine subspaces of R^3 R3, where $L_1 = U$ L1 = U and $L_2 = (2, 0, 1) + W$ L2 = (2, 0, 1) + W, for some vector subspaces U U and W W of R^3 R3. Let a basis for U U be given by $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ {(2, 0, 1), (1, 1, 0), (0, 1, 0)} and a basis for W W be given by $\{(1, 0, 1), (0, 1, 1)\}$ {(1, 0, 1), (0, 1, 1)}. Suppose there is a linear transformation $T: U \to W$ T : U → W such that $(0, 1, 0) \in ker(T)$ (0, 1, 0) ∈ ker(T), $T(2, 0, 1) = (0, 1, 1)$ T(2, 0, 1) = (0, 1, 1) and $T(1, 1, 0) = (1, 0, 1)$ T(1, 1, 0) = (1, 0, 1). An affine mapping $f: L_1 \to L_2$ f : L1 → L2 is obtained by defining $f(u) = (2, 0, 1) + T(u)$ f(u) = (2, 0, 1) + T(u), for all $u \in U$ u ∈ U. Which of the following options are true?

- [ ] $L_1 = R^3$ L1 = R3
- [ ] $L_2 = \{(x, y, z) \mid x + y - z = 1\}$ L2 = {(x, y, z) | x + y − z = 1}
- [ ] $L_2 = R^3$ L2 = R3
- [ ] $f(x, y, z) = (x - 2z + 2, z, x - z + 1)$ f(x, y, z) = (x − 2z + 2, z, x − z + 1)
- [ ] $f(x, y, z) = (x - 2z, z, x - z)$ f(x, y, z) = (x − 2z, z, x − z)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$L_1 = R^3$ L1 = R3
$L_2 = \{(x, y, z) \mid x + y - z = 1\}$ L2 = {(x, y, z) | x + y − z = 1}
$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$ f(x, y, z) = (x − 2z + 2, z, x − z + 1)

Solution:

Discussion on Option 1:Discussion on Option 1: It is given that $L_1 = U$ L1 = U and $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ {(2, 0, 1), (1, 1, 0), (0, 1, 0)} is a basis of the vector subspace U U.

The dimension of U U is

Hint

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

$L_1 = U$ L1 = U is a subspace of R^3 R3. To know whether $L_1 = U$ L1 = U is a proper subspac

R3 or not, we just have to compare their dimensions.

1 point
Now, can you see why $L_1 = U = R^3$ L1 = U = R3 ?
○ Yes
○ No
No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
Yes

Discussion on Options 2 and 3: Discussion on Options 2 and 3: It is given that $L_2 = (2, 0, 1) + W$ L2 = (2, 0, 1) + W and $\{(1, 0, 1), (0, 1, 1)\}$ {(1, 0, 1), (0, 1, 1)} is a basis for the vector subspace W W. Now to find the explicit description of L_2 L2, we have to find the explicit description of W W first.

$$W = \{a(1, 0, 1) + b(0, 1, 1) \mid a, b \in R\}$$ W =
$$= \{(a, b, a + b) \mid a, b \in R\}$$

{a(1, 0, 1) + b(0, 1, 1) ∣ a, b ∈ R} = {(a, b, a + b) ∣ a, b ∈ R}
As W W is a subspace of R^3 R3, we want to express it using the usual x, y, z x, y, z co-ordinate and the relation between them. From the above description we have, $x = a$ x = a, $y = b$ y = b, $z = a + b$ z = a + b, which implies that $z = x + y$ z = x + y. So

$$W = \{(x, y, z) \mid x + y - z = 0. \text{ where } x, y, z \in R\}$$ W =
{(x, y, z) ∣ x + y − z = 0. where x, y, z ∈ R}
1 point
Now, are you able to find the equation of L_2 L2?
○ Yes
○ No
No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
Yes

1 point
Do you agree that the sets $\{(2 + x, y, z + 1) | x + y - z = 0\}$ {(2 + x, y, z + 1) ∣ x + y − z = 0} and $\{(a, b, c) \mid a + b - c = 1\}$ {(a, b, c) ∣ a + b − c = 1} are the same?
○ Yes
○ No
No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
Yes

Discussion on Option 4 and 5:Discussion on Option 4 and 5:

Step 1:Step 1: Here, we have to find an algebraic expression for the affine map $f: L_1 \to L_2$ f : L1 → L2. To get an algebraic expression for f f, we have to find the algebraic expression which represents the linear transformation $T: U \to W$ T : U → W.

Given:

$$T(0,1,0) = (0,0,0), \quad T(2,0,1) = (0,1,1). \quad T(1,1,0) = (1,0,1)$$ T(0, 1, 0) = (0, 0, 0), T(2, 0, 1) = (0, 1, 1). T(1, 1, 0) = (1, 0, 1)

1 point

Is T T uniquely determined by the above information?

- Yes
- No

No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
Yes

Step 2:Step 2: Since $\{(0,1,0),(2,0,1),(1,1,0)\}$ {(0, 1, 0), (2, 0, 1), (1, 1, 0)} is a basis of $U = R^3$ U = R3, we can write any element $(x, y, z) \in U$ (x, y, z) ∈ U as a linear combination of $\{(0,1,0),(2,0,1),(1,1,0)\}$ {(0, 1, 0), (2, 0, 1), (1, 1, 0)}.

$$\begin{equation} (x,y,z)=a\, (0,1,0)+b\,(2,0,1)+c\,(1,1,0) \end{equation} \quad (1)$$

Apply T T on both sides:

$$T(x, y, z) = aT(0,1,0) + bT(2,0,1) + cT(1,1,0)$$ T (x, y, z) = a T (0, 1, 0) + b T (2, 0, 1) + c T (1, 1, 0)

$$T(x, y, z) = a(0,0,0) + b(0,1,1) + c(1,0,1)$$ T (x, y, z) = a (0, 0, 0) + b (0, 1, 1) + c (1, 0, 1)

To get an algebraic expression for $T(x, y, z)$ T (x, y, z), we just have to know the values of a a, b b and c c in terms of x x, y y and z z.

1 point

Now, can you find the values of a a, b b and c c using equation (1)(1)?

- Yes
- No

No, the answer is incorrect.
Score: 0
Feedback:

Accepted Answers:
Yes

Solve the above system for a, b a, b and c c.

1 point

What is the value of a a ?

- z z

○
$x - 2z$ x − 2z
○
$y + 2z - x$ y + 2z − x
○
$x - y$ x − y

No, the answer is incorrect.
Score: 0
Accepted Answers:
$y + 2z - x$ y + 2z − x

1 point
What is the value of b b ?
○
z z
○
$x - 2z$ x − 2z
○
$y + 2z - x$ y + 2z − x
○
$x - y$ x − y

No, the answer is incorrect.
Score: 0
Accepted Answers:
z z

1 point
What is the value of c c ?
○
z z
○
$x - 2z$ x − 2z
○
$y + 2z - x$ y + 2z − x
○
$x - y$ x − y

No, the answer is incorrect.
Score: 0
Accepted Answers:
$x - 2z$ x − 2z

Step 3:Step 3:

$T(x, y, z) = (0, z, z) + (x - 2z, 0, x - 2z) = (x - 2z, z, x - z)$ T (x, y, z) = (0, z, z) + (x − 2z, 0, x − 2z) = (x − 2z, z, x − z)

By definition:

$f(x, y, z) = (2, 0, 1) + T(x, y, z) \quad \text{where} (x, y, z) \in U$. f (x, y, z) = (2, 0, 1) + T (x, y, z) where (x, y, z) ∈ U.

By substituting the value $T(x, y, z)$ T (x, y, z) in the above expression, we get

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1) \quad \text{for all} (x, y, z) \in U.$$ $\mathrm{f}(\mathrm{x}, \mathrm{y}, \mathrm{z}) = (\mathrm{x} - 2\mathrm{z} + 2, \mathrm{z}, \mathrm{x} - \mathrm{z} + 1) \quad \text{for all } (\mathrm{x}, \mathrm{y}, \mathrm{z}) \in \mathrm{U}.$

Check Answers